ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

SYSTEMS ANALYSIS AND DESIGN

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Information Technology and Computer Science, Level 3. Section A: objective questions (30) · Section B: the SDLC and feasibility, fact-finding, data flow diagrams, data modelling (ERDs and the data dictionary), input and output design (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Draw diagrams neatly with a pencil and ruler, using the correct symbols.
5. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The type of feasibility that asks "Will the users accept and be able to use the new system?" is …

A technical feasibility. B economic feasibility.
C schedule feasibility. D operational feasibility.

1.1.2 In a data flow diagram (DFD), a person or organisation outside the system that sends or receives data is an …

A process. B external entity.
C data store. D attribute.

1.1.3 A context diagram shows …

A every field of the database. B the network layout.
C the screen design. D the whole system as one process with its external entities.

1.1.4 In an ERD, one customer can place many orders. This relationship is …

A one-to-one. B one-to-many. C many-to-many. D recursive.

1.1.5 A field that uniquely identifies each record in a table is the …

A foreign key. B index card. C primary key. D attribute type.

1.1.6 A check that a value falls between a lower and an upper limit is a …

A range check. B type check. C presence check. D format check.

1.1.7 The document that lists all data elements, their types, sizes and descriptions is the …

A user manual. B data dictionary.
C Gantt chart. D feasibility report.

1.1.8 Which SDLC model develops the system in small parts with frequent feedback from users?

A Waterfall B Big bang C Linear D Agile

1.1.9 A field in one table that links to the primary key of another table is a …

A foreign key. B composite key. C candidate key. D null value.

1.1.10 Converting to the new system one department at a time is …

A direct conversion. B parallel conversion.
C phased conversion. D pilot conversion.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 In a DFD, data may flow directly from one data store to another without a process.

1.2.2 Observation lets the analyst see how work is actually done.

1.2.3 A prototype is a working model used to get feedback from users early.

1.2.4 Economic feasibility compares the costs and the benefits of a system.

1.2.5 A questionnaire is the best method to get detailed answers from one manager.

1.2.6 Normalisation reduces repeated (redundant) data in a database.

1.2.7 A good input screen should have as many colours and fonts as possible.

1.2.8 A verification check asks the user to enter a password twice.

1.2.9 Every process in a DFD must have at least one input and one output data flow.

1.2.10 The systems analyst writes the final program code in all cases.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	A person or thing about which data is stored, e.g. STUDENT	A	pilot conversion
1.3.2	A property of an entity, e.g. StudentNumber	B	end user
1.3.3	Where data is kept in a DFD, e.g. a file or table	C	scope
1.3.4	Transforms incoming data into outgoing data in a DFD	D	stakeholder
1.3.5	Person who will use the new system every day	E	report (output)
1.3.6	Anyone with an interest in the system, e.g. managers, users, clients	F	process
1.3.7	Trying out the new system in one branch first	G	presence check
1.3.8	The boundary of what the project will and will not include	H	attribute
1.3.9	Check that a field has not been left empty	I	entity
1.3.10	Printed or on-screen result produced by the system	J	data store

(10)

TOTAL SECTION A: [30]

SECTION B

Use the following case study to answer QUESTIONS 2 to 5.

Case study: Ubuntu Community Library
The Ubuntu Community Library lends books to about 1 200 members. Members register by giving their ID number, name, address and cellphone number, and they receive a membership card. A member may borrow up to three books for 14 days. The librarian writes each loan in a large book: member number, book number, date borrowed and date due. When books are returned late, a fine of R2 per day is charged. Books often get lost because nobody knows who has them, and the librarian spends hours each month finding overdue books. The library committee has a small budget from the municipality and wants a computer system to manage members, books, loans and fines, and to send SMS reminders when books are overdue.

QUESTION 2: THE SDLC AND FEASIBILITY

2.1 Identify THREE problems with the current manual system. (3)

2.2 Write TWO objectives for the new system. (2)

2.3 Explain the FOUR types of feasibility the analyst must consider, and apply EACH to the library. (8)

2.4 Recommend a conversion method for the library and justify your choice. (2)

[15]

QUESTION 3: FACT-FINDING

3.1 Recommend a suitable fact-finding technique to gather information from EACH of the following, and give a reason: (a) the librarian; (b) the 1 200 members. (4)

3.2 Write THREE questions that the analyst could ask the librarian. (3)

3.3 Give ONE disadvantage of observation as a fact-finding technique. (1)

3.4 The analyst plans to show the librarian a prototype of the loans screen. Give THREE advantages of prototyping. (3)

[11]

QUESTION 4: DATA FLOW DIAGRAMS

4.1 Draw a context diagram for the new library system. Show at least TWO external entities and FOUR labelled data flows. (8)

4.2 Draw a level-0 DFD for the process "Return a book". Include the processes for recording the return and calculating the fine, at least TWO data stores and the member as an external entity. (8)

4.3 Name the FOUR symbols used in a DFD (Gane and Sarson or Yourdon). (4)

[20]

QUESTION 5: DATA MODELLING AND DESIGN

5.1 Identify THREE entities for the library database. For EACH entity give a suitable primary key and TWO other attributes. (6)

5.2 Draw an entity relationship diagram (ERD) showing the relationships between MEMBER, LOAN and BOOK. Show the cardinality. (5)

5.3 Complete a data dictionary entry for the field MemberCellphone: field name, data type, size, format and a validation rule. (5)

5.4 Give FOUR guidelines for designing a user-friendly input screen to capture a new loan. (4)

5.5 Design the content of the SMS reminder that the system sends for an overdue book. Indicate which data comes from the database. (2)

5.6 Calculate the fine for a book that was due on 3 March and returned on 17 March. (2)

[24]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER